COVID-19 TRACKER

a minor project report, submitted in partial fulfillment of the requirements for the award of the degree of

B. Tech.

In

Information Technology

By

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(2017IMG-061)



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CANDIDATE/S DECLARATION

I hereby certify that the work, which is being presented in the report/thesis, entitled **COVID-19 TRACKER ,** in partial fulfillment of the requirement for the award of the degree of **Bachelor of Technology** and submitted to the institution is an authentic record of my/our own work carried out during the period *July - 2020* to *October – 2020* under the supervision of Prof. Manoj Patwardhan. I also cited about the text(s)/ figure(s)/ tables(s) from where they have been taken.

The matter presented in this thesis has not been submitted elsewhere for the award of any other degree of diploma from any institutions. (This is only applicable for B. Tech. (IT))

Date: Signature of the candidate

This is to certify that the above statement made by the candidate is correct to the best of my/our knowledge

Date: Signature(s) of the supervisor(s)

The Ph.D. Viva-Voce examination of Yash Agrawal, Research Scholar has been held on _____

Signature of Research Supervisor	Signature of External examiner	Signature of Chair person of Research Committee

ABSTRACT

COVID-19 spread in the entire world very swiftly. Medical professionals and others have differently speculated about the spread and eradication of the disease with respect to time

In this study an attempt has been made to scientifically project the spread of the disease based on previous available data

Our approach is based on the curve fitting by statistical tools to find out the average rate of growth of COVID-19 virus and project it further to find out the future spread of the disease

The proposed system is based on some assumptions apart from real time application.

A computer program has been written in python to carry out data analysis and forecast

The results are shown to verify that model tries to help in finding the approximate growth rate of COVID-19 disease in real time environment

Keywords:

Corona Virus, Statistical Model, Growth Rate, OLS (ordinary least square distance)

Acknowledgements

I am highly indebted to Prof. MANOJ PATWARDHAN, and am obliged for giving me the autonomy of functioning and experimenting with ideas.

I would like to take this opportunity to express my profound gratitude to him, not only for his academic guidance, but also for his personal interest in my project and constant support coupled with confidence boosting and motivating sessions which proved very fruitful and were instrumental in infusing self assurance and trust within me.

The nurturing and blossoming of the present work (up to whatever extent) is mainly due to his valuable guidance, suggestions, astute judgment, constructive criticism and an eye for perfection.

My mentor always answered myriad of my doubts with smiling graciousness and prodigious patience, never letting me feel that I am novice by always lending an ear to my views, appreciating and improving them, and giving me a free hand in my project.

It's only because of his overwhelming interest and helpful attitude, the present work has attained the stage it has.

Finally, I am grateful to my institution, BTP convener, colleagues and my family whose constant encouragement served to renew my spirit, refocus my attention and energy and helped me in carrying out this work.

Last but not the least, I am grateful and thankful to the readers who will be kind enough to spare some time from their busy schedule to read it and giving their in valueable Criticism/Suggestions for improvement which will be welcomed, so that this work continues to grow beyond its destiny

Yash Agrawal

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- Csv File (link for same is in references ,please see last page)

- (date of cases(1 jan 2020 to 15 oct 2020), total confirmed cases, daily cases

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Please See chapter 6

CHAPTER 1

Introduction

COVID-19 TRACKER (this project) has been made as simple as possible in comparison to various versions of COVID-19 Tracker application(s) which can be found on internet.

Everyday we find Media reports, Responsible authorities (W.H.O., Government) and Medical experts speculating the spread and extinction of the disease. They have been doing this without any scientific/ mathematical approach instead vaguely speculating by subjectively looking at other country's data and experiences. It is known to everybody that all such speculations have proven myth.

In this study past data is used to analyze the growth of corona virus confirmed cases, and a mathematical model is fit, a computer program written to analyze and predict the future confirmed cases. Model is developed by method of best fit curve (an n degree polynomial).\visualized by graphs (mathematical diagrams). An overview of COVID-19 cases amongst countries all over the world has been given with help of pie charts

Several assumptions have been made to make the study as simple as possible viz. mutations in corona virus, degree of acceptability of prevention measures by the masses, 2nd Corona virus wave etc. Many other scientists and researchers have carefully kept such assumptions in mind but still their forecast has been null and void, as we all know.

Since nearly every Scientist(s), Institutions are continuously improving their forecast of covid-19 cases (which seems 100% correct in its own), just because corona virus is unique in its own and since past experience suggests that historic pandemic(s) have persisted at least 1yr and at most 3 yr, according to this model COVID-19 pandemic will persist at least 1yr which fosters its correctness

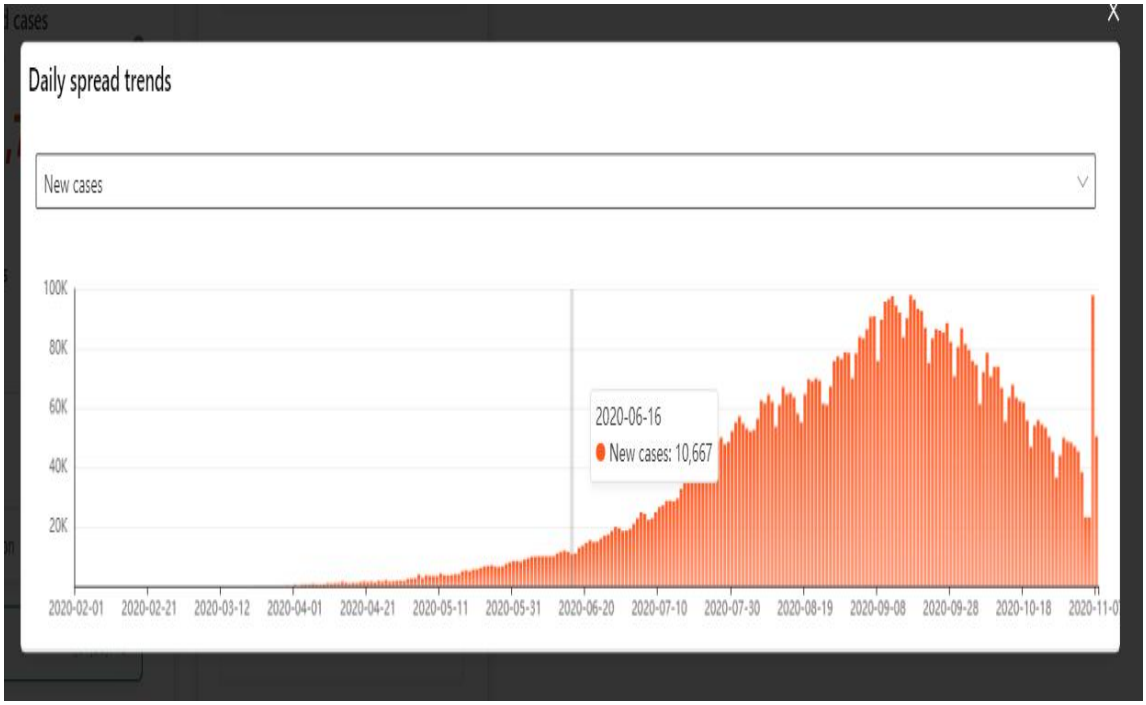
CHAPTER 2

Literature Survey

0. Various Literature track COVID-19 trajectory, make a model by analyzing from different point of views, e.g. age structured impact on COVID-19, Effect of lockdowns on COVID-19 and many have forecasted when and how much lockdowns in future can be optimum to control pandemic. But all this is highly subjective and any relevant conclusions cannot be expected

Some of recent literature survey is reproduced below:

1. **COVID-19 tracker by Bing. https://bing.com/covid/local/india**



(Where are the forecasts??)

2. **Medical director AIIMS on COVID-19 spread https://www.republicworld.com/india-news**



3. https://www.researchgate.net/publication/340209224_Age-structured_impact_of_social_distancing_on_the_COVID-19_epidemic_in_India_Updates_at_httpsgithubcomrajeshrinetpyross
   In above article/study, a modified version of general epidemic model has been used:. Here Infected persons only are responsible for spreading of disease by contact (which are dependant on age-group), and a differential equation is fitted. . After solving differential equation, a model is fitted and is used to predict other cases

   Asymptomatic people have not been covered although we know now that a sizeable number of them are also responsible for spread.

CHAPTER 3

Requirements Analysis and Specification

System Requirements for project:

A simple windows 10 pc with Anaconda (for data analysis using python) and MATLAB/Octave/SCILAB installed (probably latest version) and internet connection.

Specification of system:

64bit pc

CHAPTER 4

System Design, Project Description and Methodology

System Design:

input/raw data→model→output/information

Project description:

Tracking COVID-19 growth.

Assumptions: no mutation in corona wave and others, that is corona virus is one of the simplest virus till date.

Methodology:

Best fit curve:

Let curve be polynomial of degree n.

We apply Ordinary Least Square distance(OLS). And partially differentiate it with respect to coefficients(1 at time) to get n+1 linear system of equations in n+1 variable(the coefficients)

Solved the system by matrix algebra by software like MATLAB/octave if not possible in python

However the code's logic part is shown in screenshot in ch6

(for understanding best fit curve more, please check for link in last page)

CHAPTER 5

Implementation Results

After solving the unknown coefficient vector, we fit these coefficients into assumed polynomial equation, and hence get model, in which I/p data when fitted yields o/p values.

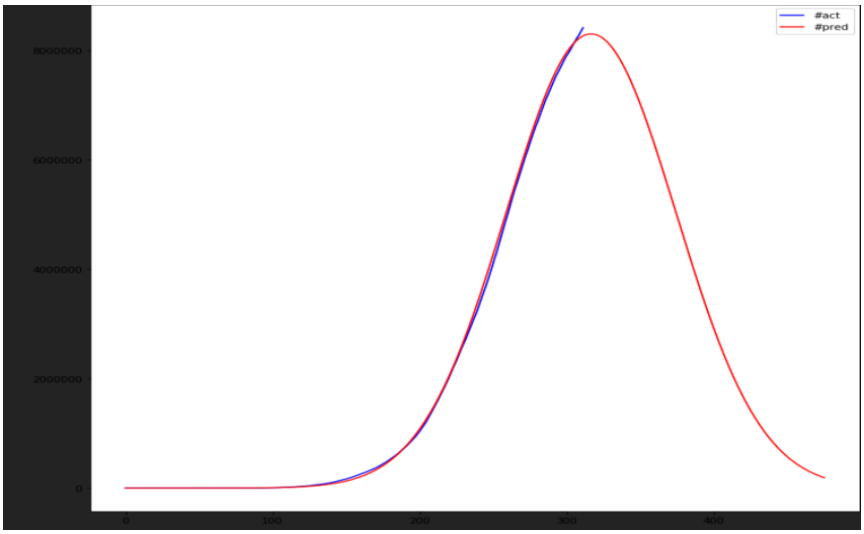
India has maximum recovery cases, and has 2nd largest total cases, U.S. has maximum total cases, as well as maximum death cases, and less recovery rate, indicating that U.S. is worst hit by pandemic

Of the 2 models applied, first one predicts, pandemic will end after this December, while second one predicts after March.

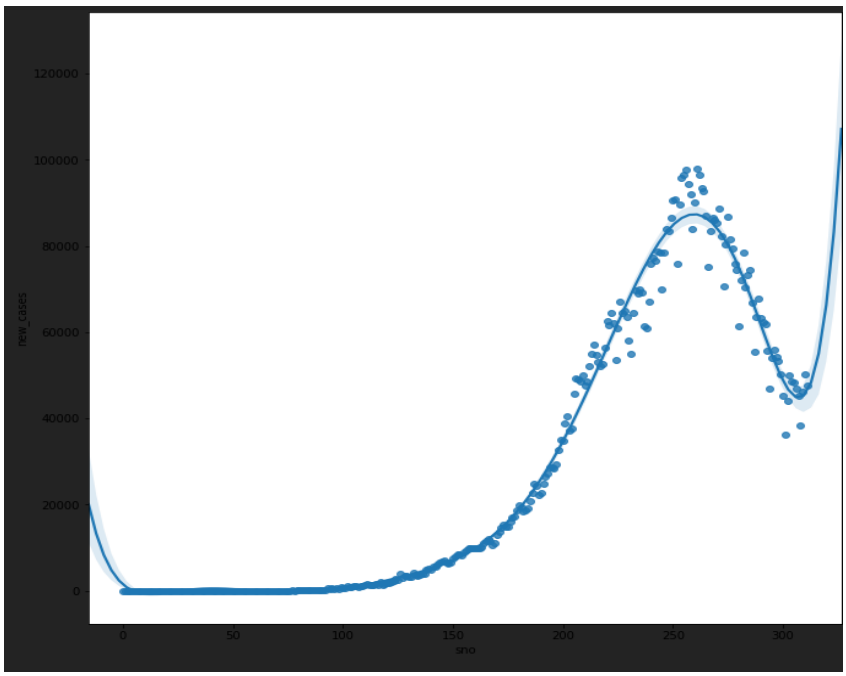
1st model(polynomial curve) looks more accurate than 2nd model(normal curve fit).

CHAPTER 6

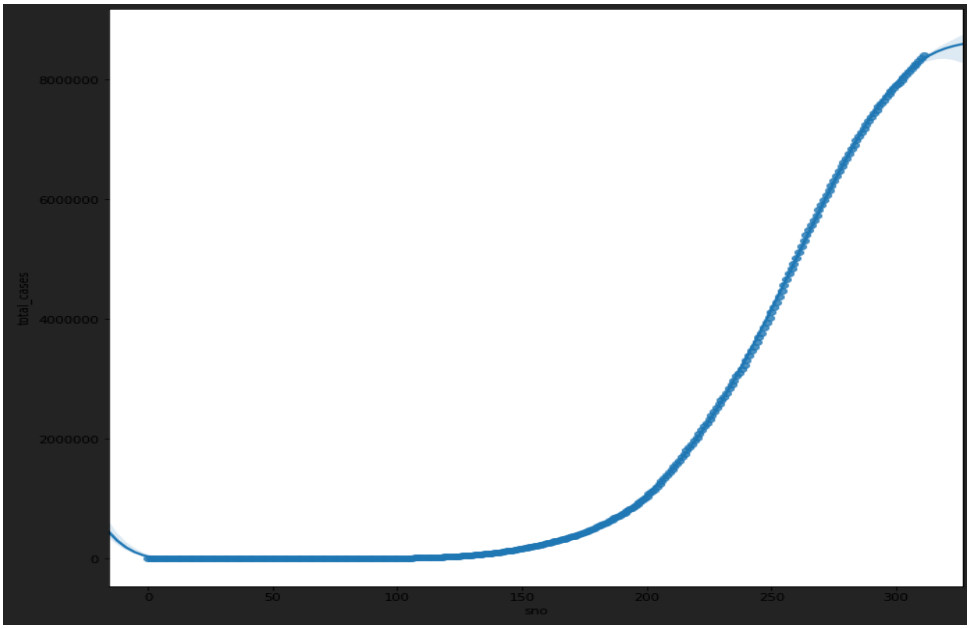
Screenshots of Project with Descriptions



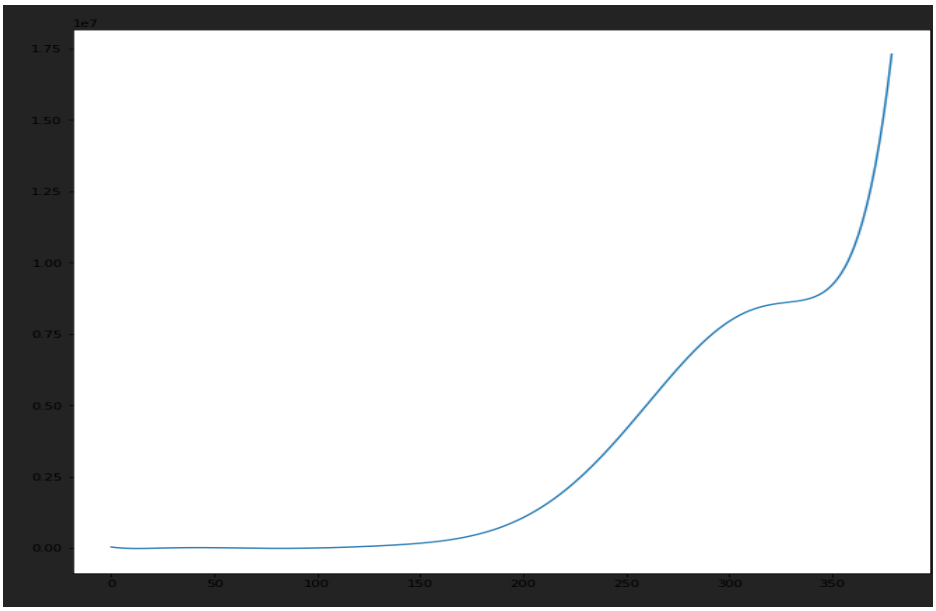
model 2 fitting normal curve in total_cases(blue=> actual, red=> predicted)



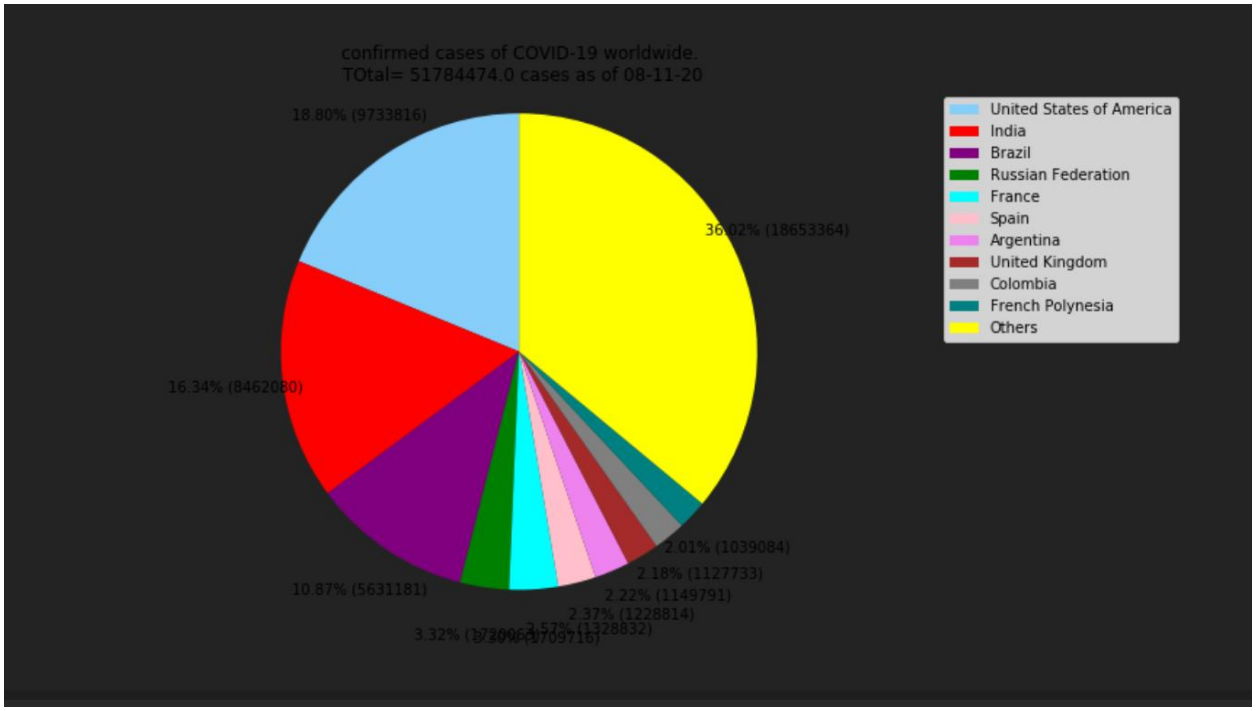
(new cases vs date, scattered->actual, continuum->model, y-axis: new cases, x-axis: date, new cases soon to reach its minimum, cases may rise in winters in conformation to HONBL DR. GULLERIA)



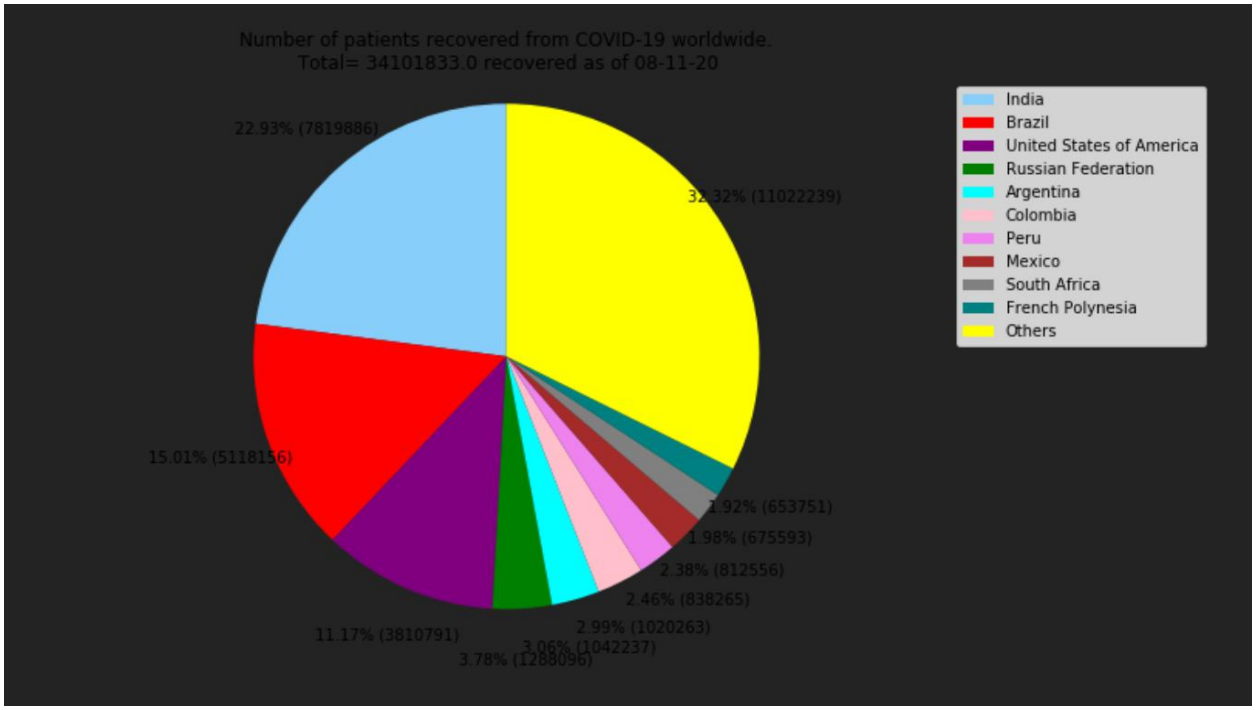
(continuum: model, scattered : actual, y-axis: total cases, x-axis: dates)



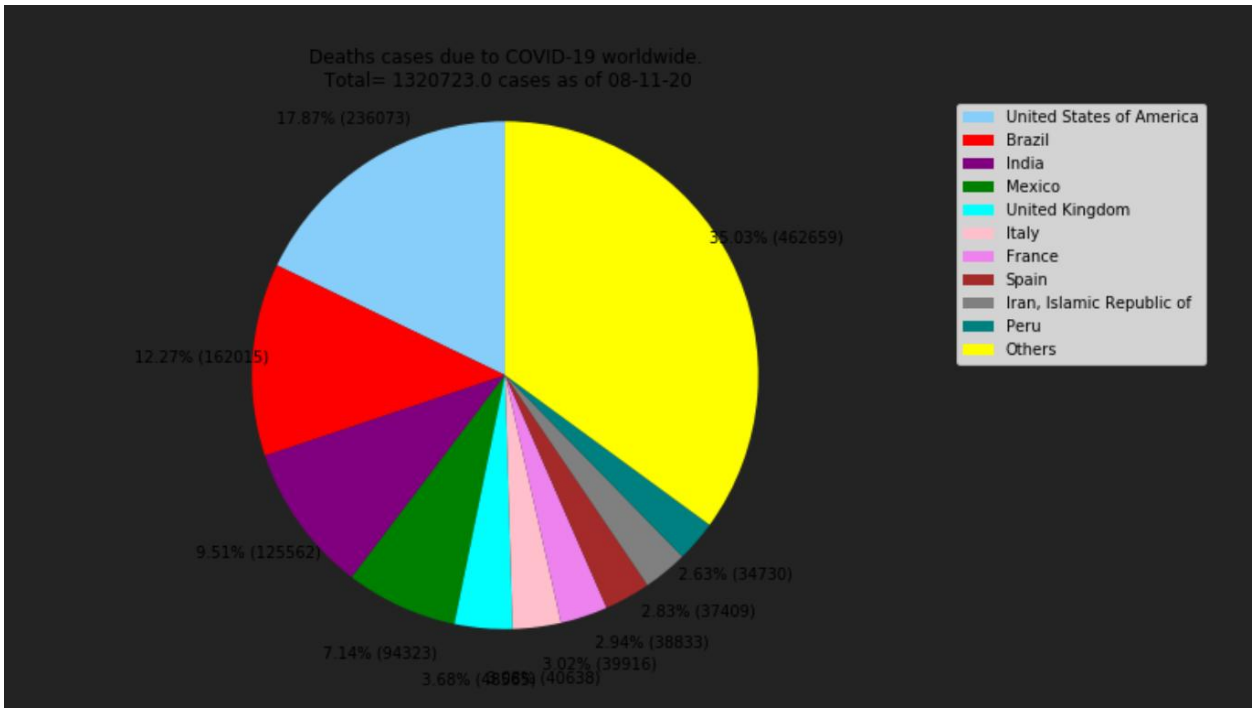
(the model extended for future dates , possibility of covid-19 2nd wave, rising cases in winters in conformation with HONBL Dr. GULLERIA)



Comparison of countries (total cases){RED COLOR}



Comparison of countries(recovered cases)[India has most recovery rate and cases as claimed by Government]{SKY BLUE}



Comparison of countries(deaths){PURPLE COLOR}

```
In [17]:
df_y=df['new_cases']            # the y axis values
df_x=df['datesno']              # the x axis values(entires e.g. 31jan2020=>0, 29 Feb=>29, etc.. )
deg=12                          # highest degree of polynomial equation to be fitted
ps=[]                           # y*(x^r)
l=len(df_y)                     # number of records
for i in range(deg+1):          # column vector (b) in AX=B
    ps.append(0)                # initilaize B=0
s=[]                            # matrix A in AX=B
for i in range(2*deg+1):        # initialize to 0 EVERY ENTRY OF MATRIX
    s.append(0)
for i in range(l):              # time to prepare/ calculate the entries for A, B
    y=df_y.iloc[i]              # one value of dependent variable(new_cases) at time
    x=df_x.iloc[i]              # one value of independent variable(total_cases) at time
    for j in range(deg+1):      # appropriately fill B by appropriate calculations
        ps[j]=ps[j]+y*(x**j)
    for j in range(2*deg+1):    # appropritate fill A by apporpriate calculations
        s[j]=s[j]+(x**j)
A=[]                            # set A
B=[]                            # set B
for i in range(deg+1):
    SA=[]
    B.append(ps[deg-i])
    for j in range(deg+1):
        SA.append(s[2*deg-j-i])
    A.append(SA)
A=np.array(A)                   # now A , B is hopefully ready to be processed to solve for coefficient matrix
B=np.array(B)
```

(the logic code of fitting model, rest of code(not shown in fig) is play of presenting plots to help to visualize.

```
In [22]:
def update_csv(file, df):
    global today
    if (os.path.exists(file)):
        original_countries = pd.read_csv('confirmed.csv')
        current_df = pd.read_csv(file)
        current_df[today] = pd.DataFrame(columns = [today])
        today_data = []
        yesterday = str((datetime.strptime(today, "%d-%m-%y") - timedelta(days = 92)).strftime("%d-%m-%y"))

        for country in original_countries.iloc[:, 0]:
            if (country in list(df.loc[:, 'Country'])):
                today_data.append(int(df[df['Country'] == country][today]))
            else:
                today_data.append(current_df.loc[current_df['Country'] == country][yesterday].item())
        current_df[today] = pd.Series(today_data)
        current_df.to_csv(file, index=False)
    else:
        countries = requests.get("https://api.covid19api.com/countries").json()
        countries = sorted([i['Country'] for i in countries])
        country_df = pd.DataFrame(data = countries, columns = ['Country'])
        country_df[today] = df[today]
        country_df.to_csv(file, index=False)

In [23]:
data = requests.get("https://api.covid19api.com/summary").json()
today = str(datetime.strptime(data['Countries'][0]['Date'][:10], "%Y-%m-%d").strftime("%d-%m-%y"))
global_data = data["Global"]
print(f"\nGlobal status as of {today}")
print(f"Total cases: {global_data['TotalConfirmed']}")
print(f"New cases since Yesterday: {global_data['NewConfirmed']}")
print(f"Total Deaths: {global_data['TotalDeaths']}")
```

the logic part of program is data handling, rest is play of presentations

date	actual	model1	%diff	model2	%diff
1-Nov-20	8184082	8187321	-0.04	8164034	0.24
2-Nov-20	8229313	8222409	0.08	8188582	0.49
3-Nov-20	8267623	8255667	0.14	8210735	0.69
4-Nov-20	8313876	8287109	0.32	8230473	1.00
5-Nov-20	8364086	8316757	0.57	8247779	1.39
6-Nov-20	8411724	8344635	0.80	8262636	1.77
7-Nov-20	8462080	8370776	1.08	8275031	2.21
8-Nov-20	8510219	8395217	1.35	8284954	2.65

Comparison of model 2(normal curve) vs model 1 (a general polynomial curve){model 1 explains better}

CHAPTER 7

Conclusions and Future Scope

// As per 4th[d] graphical sheet pandemic in India is forecasted to grow further in winters in confirmation with AIIMS director Dr. RANDEEP GULERIA and further growth indicating COVID-19 2[nd] wave in early 2021. //

Future scope:

There are multiple possibilities of further developing this study.

1. The data used for this Mathematical model *incorporates and speaks in itself* about all dimensions of known and unknown COVID-19 properties and their effect on the population viz. climate factors , morphology and behavior of the virus, diagnosis and treatment facilities, implementation of prevention measures etc.
   Hence this kind of integrated approach can be implemented for any country of the world making this study *universally applicable*.
2. This can be *customized* by generating a user friendly webpage to accept date wise data and generate *online forecast.*

REFERENCES

https://medium.com/tech-iiitg/create-your-own-covid-19-tracker-and-data-visualiser-with-pandas-and-matplotlib-in-python-a1c6e3dbd378

(COVID-19 comparison among countries in world)

https://drive.google.com/drive/u/0/my-drive

(the dataset (btp.csv))

Thank You